

Bounded multiplicative dynamics govern axonal conduction slowdown

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ABSTRACT Conduction velocity slows along axons, yet the right-skewed distribution of the terminal-to-initial velocity ratio $\rho = v_{\text{end}}/v_{\text{start}}$ has length-invariant mean and variance. A bounded multiplicative framework is introduced in which local geometric and kinetic factors compound proportionally but only within a finite distal domain set by termination conditions that saturate the effective multiplicative depth. The model accounts for the observed stability of the slowdown distribution across lengths and yields discriminating experimental signatures. More generally, it suggests that robust function can emerge through constrained variability rather than structural uniformity.

WHY IT MATTERS In spatially extended biological systems, local variations often accumulate along trajectories, commonly in a multiplicative manner: what is observed downstream reflects the preceding state multiplied by local geometric, kinetic, or material factors. Axons provide a concrete case. Conduction velocity is shaped by heterogeneities along the axonal trajectory, including geometry, membrane properties, and excitability, leading to progressive slowdown as electrical signals travel. Yet, contrary to what unbounded accumulation predicts, the slowdown distribution remains stable across branch lengths. The axonal data motivate the model offered here: a multiplicative framework in which effective compounding depth is limited by the distal termination. More generally, the framework suggests that functional invariance may depend less on structural uniformity than on variability constrained by global boundary conditions.

INTRODUCTION

Under most physiological activity regimes, including spontaneous network activity and low-to-moderate stimulation rates, axons function as robust conducting structures, reliably transmitting action potentials across distances and levels of axonal branching.^{1–7} This robustness does not imply trivial dynamics. Axonal propagation is shaped by geometry, membrane properties, and boundary conditions, and classical cable-theoretic analyses show that conduction velocity is sensitive to axial load, diameter changes, and sealed-end terminations.^{8–10} These effects are generally interpreted as modulations of a fundamentally reliable signaling process rather than as transformations of representational content conveyed by the spike itself.

Experimental and computational studies have long noted that axonal inhomogeneities can induce progressive slowing as spikes propagate away from their site of initiation.¹¹ The slowdown exhibits a stable statistical organization when examined across large axonal ensembles.¹² As demonstrated in [Figure 1](#), individual velocity profiles display a gradual, predominantly monotonic deceleration along branches. The ratio between terminal and initial velocities, $\rho = v_{\text{end}}/v_{\text{start}}$, is right skewed and well described by a log-normal distribution; across the full set of branches, the typical slowdown is about 30%.¹² Within the measured range (0.6–4.6 mm), there is no statistical evidence for length dependence of ρ , nor for broadening of the $\ln \rho$ distribution. These observations suggest a compounded, yet bounded, effect of many small, local influences.

A minimal model is introduced, grounded in cable-theoretic excitability, in which distal boundary constraints impose a terminal multiplicative prefactor that is a function of distance to the tip. This boundary-anchored contribution dominates the net effect

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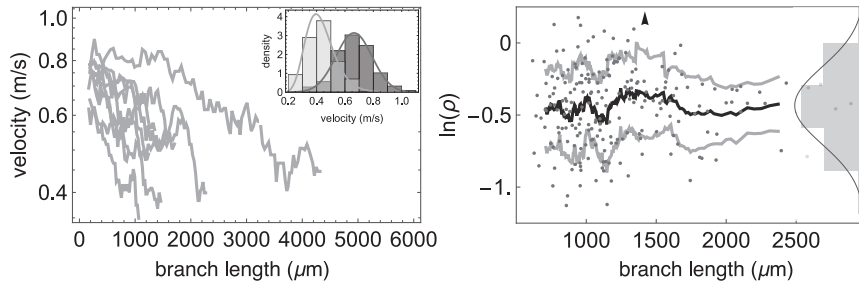


Figure 1. Statistical organization of spike propagation along axons (adapted from Cohen and Marom¹²). Left: representative log-linear conduction-velocity profiles for axons of different lengths. Inset (left): histograms of v_{start} (dark gray) and v_{end} (light gray) for branches ($n = 214$) selected to originate near the somatic spike-initiation zone by requiring an initial time of arrival (TOA) of at most 200 μs relative to the triggering somatic spike (corresponding to an origin within $\sim 130 \mu\text{m}$ given $\bar{v}_{\text{start}} = 0.663 \text{ m s}^{-1}$). Right: scatterplot of $\ln(\rho)$ versus branch length for

the same $n = 214$ subset, where $\rho = v_{\text{end}}/v_{\text{start}}$ denotes the slowdown ratio (plotted up to 3 mm for visual clarity), overlaid with a rolling mean and rolling standard-deviation bands (± 1 SD of $\ln(\rho)$; window size = 20). One point lies above the displayed y range; it is retained in all analyses and indicated by an arrow. Across the full set of 431 branches, slowdown shows no systematic dependence on length (Pearson $r = 0.018$, $p = 0.71$; Spearman $r_s = 0.044$, $p = 0.36$), and dispersion shows no evidence of length-dependent broadening (Breusch-Pagan test on residuals from a linear regression of $\ln(\rho)$ on length: $p \approx 0.56$).¹² The image displays this near-soma $n = 214$ subset, within which the same absence of length dependence holds (Pearson $r = 0.07$, $p = 0.30$; Spearman $r_s = 0.08$, $p = 0.26$; Breusch-Pagan test on residuals from a linear regression of $\ln(\rho)$ on length: $p = 0.12$). Inset (right): distribution of $\ln(\rho)$ for this near-soma $n = 214$ subset, with a normal fit ($\mu = -0.44$, $\sigma = 0.27$); the slightly more negative mean than the full-set value reflects the distal-origin restriction.

of the multiplicative updates and yields slowdown statistics that are effectively independent of total trajectory length. The proposed model resembles transport toward a distal boundary condition: regardless of upstream distance, approaching the boundary imposes a boundary-set deceleration. Any axonal terminal realization that reduces the local safety factor (conduction margin) over a finite coupling depth can limit propagation via structural, kinetic, or mixed mechanisms,¹³ including involvement of intra-axonal organelles (e.g., Castelfranco and Alcamì¹⁴). The formulation proposed here yields realization-informative signatures, that is, signatures that indicate which class of distal mechanism is operative, because distinct distal mechanisms differentially modulate both perturbation-induced changes in the expectation (mean) $\mathbb{E}[\ln \rho]$ and the effective multiplicative depth.

MULTIPLICATIVE SLOWDOWN AND THE FAILURE OF UNBOUNDED ACCUMULATION

The compounded influences are formalized by expressing conduction velocity along a branch as a recursive product of local multiplicative factors. Let v_n denote the velocity in segment n , and let $\kappa_n \in [0, \infty)$ be a dimensionless factor representing the local geometric or kinetic modulation of velocity over that increment. Then, $v_{n+1} = \kappa_n v_n$, so that over N segments, the terminal velocity satisfies

$$v_{\text{end}} = v_{\text{start}} \prod_{i=1}^N \kappa_i. \quad (1)$$

“Segment” denotes a coarse-grained spatial increment over which local proportional modulation of the source-to-load balance between regenerative

drive and downstream electrical load is summarized by a single multiplicative factor κ_i . It is not intended as an anatomical or electrotonic unit, and its scale is a modeling resolution parameter.

Formally, this is a one-dimensional multiplicative propagation scheme, in which transport along a path is described as a fixed reference scale multiplied by a product of local factors that represent successive inhomogeneities, local geometric and kinetic influences that rescale spike conduction velocity along the axon. Writing the end-to-start ratio as $\rho = v_{\text{end}}/v_{\text{start}}$ gives

$$\ln \rho = \sum_{i=1}^N \ln \kappa_i, \quad (2)$$

consistent with a right-skewed, approximately log-normal distribution for ρ . This log normality of ρ is assessed directly from the near normality of $\ln \rho$ (Figure 1, right)¹²; it is not (and in fact, cannot be) inferred from the marginal forms of v_{start} and v_{end} .

Note that the above multiplicative scheme entails a correlation of the first and second moments of $\ln \rho$ with travel distance. This, however, does not align with empirical observations (Figure 1, right). Thus, local geometric and biophysical influences might still modulate propagation proportionally, such that $\rho = \prod \kappa_i$, compounding along the trajectory, but the overall mean and variability of slowdown remain constrained. This constrained multiplicative structure provides the basis for the formal treatment offered below.

BOUNDARY-ANCHORED COARSE GRAINING

The purpose of this coarse graining is to replace many unresolved distal details by a single effective boundary factor.

In principle, conduction along a sufficiently extended axon could reflect contributions from two spatial domains. Far from a sealed or tapering termination, velocity could remain approximately constant or drift gradually in a way that need not be strictly multiplicative. If such an upstream contribution accumulated with distance, one would expect the terminal slowdown ratio $\rho = v_{\text{end}}/v_{\text{start}}$ to depend systematically on branch length. Yet, at least across the measured range of lengths described in Cohen and Marom¹² and Figure 1, the mean of $\ln \rho$ remains effectively invariant, and no gross broadening of the slowdown distribution is evident in the aggregate data. Absent fine-tuned cancellation, the most economical interpretation is that any proximal changes in velocity do not contribute appreciably to the statistics of ρ and can be absorbed into the branch-specific reference scale v_{start} . These considerations motivate restricting the present analysis to a bounded distal contribution near an effective termination, and the toy model below focuses on the corresponding finite multiplicative depth within the distal coupling zone set by the terminal boundary. Here, Λ denotes the effective active range over which distal constraints can influence regenerative propagation; it is distinguished from the passive electrotonic length λ , which delimits linear subthreshold locality.

Thus, the boundary factor $\eta(\Lambda)$ is defined at the level of an effective realization rather than a microscopic mechanism; it summarizes the distal renormalization of the local source-to-load balance that sets v_{end} . It may arise from structural load/impedance effects, from excitability effects, or from combinations of these. The sealed termination here principally sets the scale Λ over which these distal influences remain coupled to the terminal ratio rather than itself fixing the sign of the bias. The normalized end-to-start ratio can then be written schematically as

$$\rho \equiv \frac{v_{\text{end}}}{v_{\text{start}}} = \eta(\Lambda) \prod_i \kappa_i. \quad (3)$$

The ratio is dimensionless by definition. Here, $\eta(\Lambda)$ denotes the effective distal boundary contribution, whereas the κ_i encode residual spatial heterogeneity within the boundary-coupled distal segment.

The effective distal bias can be conceptualized as comprising two distinct components,

$$\eta(\Lambda) = \eta_{\text{structure}}(\Lambda) \eta_{\text{kinetic}}(\Lambda), \quad (4)$$

where $\eta_{\text{structure}}$ summarizes effectively time-independent distal constraints on the source-to-load balance,^{8–10,13} including taper and impedance mismatch toward the termination, and other struc-

tural or nearly static spatial factors (for example, diameter variation, branching, axial resistivity, or fixed gradients in channel expression). Concrete structural realizations of this term are illustrated in Figure S1. In contrast, η_{kinetic} summarizes history- and voltage-dependent adaptive processes that reduce excitability reserve.^{6,10,15–18} These include availability-dependent kinetics, defined here as changes in the fraction of channels available for activation, for example through inactivation, accommodation-like losses, or channel-gating biochemical modulation. A Hodgkin-Huxley-based illustrative proxy and waveform signature of reduced availability are shown in Figure S2.

Equation 4 is a log-additive bookkeeping device, not an assertion that structural and kinetic contributions are mechanistically independent. Any nonseparable coupling can be absorbed into the Λ dependence of the two factors or represented explicitly by an interaction term. This (nonunique) decomposition is chosen to keep the model implementation agnostic while enabling predictions that discriminate among realization classes. Empirically, under typical conditions, $\eta(\Lambda) < 1$, so the distal boundary acts as a net decelerating bias that aggregates load/impedance and excitability-reserve effects into a single renormalizing operator.

BOUNDED MULTIPLICATIVE DEPTH AND LENGTH-INVARIANT DISPERSION

A simple analytic construction clarifies how the distal factor $\eta(\Lambda)$ can preserve the log-normal form of slowdown while suppressing the length dependence of $\text{Var}[\ln \rho]$ characteristic of a multiplicative propagation scheme. Let

$$\rho = \eta(\Lambda) \prod_{i=1}^{n(L)} \kappa_i, \zeta_i \equiv \ln \kappa_i, \quad (5)$$

where L denotes distance measured from the sealed termination back toward the soma and ζ_i are local log increments with mean μ and variance σ^2 . Introducing ζ_i linearizes multiplicative compounding, so that $\ln \rho$ can be analyzed using standard results for sums of weakly dependent increments.

In a passive one-dimensional conductor, the number of effectively independent increments $n(L)$ grows linearly with distance, yielding

$$\text{Var}[\ln \rho] = n(L)\sigma^2. \quad (6)$$

This is the source of length-dependent variance characteristic of multiplicative one-dimensional

propagation schemes. In an axon, however, the sealed or tapering termination imposes a boundary condition whose influence decays with distance from the tip with a characteristic scale Λ . Beyond that scale, additional distal cable couples only weakly to the terminal ratio ρ through the distal boundary condition, so the effective number of statistically independent multiplicative contributions that can influence the terminal slowdown saturates rather than growing linearly. This saturation concerns sensitivity of the terminal ratio statistic $\rho = v_{\text{end}}/v_{\text{start}}$ to boundary-coupled fluctuations; upstream heterogeneity may still contribute to absolute travel time without remaining coupled to the sealed-end source-to-load margin that determines v_{end} .

A minimal coarse-grained representation for this boundary-limited coupling is a saturating effective depth,

$$n_{\text{eff}}(L) = n_{\infty}(1 - e^{-L/\Lambda}), \quad (7)$$

where Λ is an effective coupling length (not necessarily identical to the passive electrotonic λ) and n_{∞} is the saturated depth. This formulation does not assume that local multiplicative factors differ intrinsically along the axon. Rather, it reflects that only fluctuations that remain coupled to the distal boundary can influence the terminal ratio, and this influence is strongest when propagation is near threshold, where small variations in local conductances or impedance produce disproportionately large changes in regenerative balance. The saturation of n_{eff} is therefore a statement about coupling and sensitivity, not about the absence of heterogeneity elsewhere. The resulting slowdown can be written as

$$\ln \rho = \ln \eta(\Lambda) + \sum_{i=1}^{n_{\text{eff}}(L)} \zeta_i. \quad (8)$$

Here, $n_{\text{eff}}(L)$, though continuous, is used in a coarse-grained sense. Thus,

$$\text{Var}[\ln \rho] = n_{\text{eff}}(L)\sigma^2, \quad (9)$$

with

$$\text{Var}[\ln \rho] \xrightarrow{L \rightarrow \infty} n_{\infty}\sigma^2, \quad (10)$$

a finite constant. The mean slowdown, by contrast, continues to depend on $\ln \eta(\Lambda)$ and on the saturated depth through $n_{\infty}\mu$. This toy model shows that saturation of the effective multiplicative depth $n_{\text{eff}}(L)$ suppresses the growth of $\text{Var}[\ln \rho]$ by limiting the

number of independent multiplicative perturbations that remain coupled to the terminal statistic, whereas deterministic boundary factors embedded in $\eta(\Lambda)$ (including the components $\eta_{\text{structure}}$ and η_{kinetic}) primarily shift $\mathbb{E}[\ln \rho]$ through $\ln \eta(\Lambda)$ without setting the dispersion plateau. The result is a bounded multiplicative process with log-normal form and length-invariant dispersion, consistent with the empirical slowdown observed in axons.

A Monte Carlo comparison illustrates that removing saturation in $n_{\text{eff}}(L)$ produces length-dependent broadening of $\text{Var}[\ln \rho]$ over the measured range, whereas a bounded effective depth reproduces the observed length-invariant dispersion (Figure 2).

ALTERNATIVE MODELS

The data narrow the space of plausible mechanisms; it is therefore useful to consider several alternative realizations and the specific empirical requirements they do, or do not, satisfy. The empirical concurrence of a right-skewed distribution of ρ , weak dependence of $\text{Var}[\ln \rho]$ on branch length, and mean and median ρ values near 0.7 that remain effectively length-invariant, appears to require some bounded accumulation principle. In the present formulation, that requirement is expressed phenomenologically as a distal boundary constraint: slowdown accumulates multiplicatively but only up to an effective distal limit, so that progressive deceleration does not translate into unbounded broadening or continued drift of central tendency with distance.

One possible realization of this logic is a discrete weak-link or branch-point picture, in which

$$\rho \approx \prod_{k=1}^B \kappa_k, \quad (11)$$

where each $\kappa_k < 1$ represents a local slowing event and B is the effective number of such events encountered along the trajectory. If B is capped or varies only weakly across branches, then multiplicative compounding can generate both a stable central tendency and a right-skewed distribution without strong dependence on branch length. In that sense, the weak-link picture may be viewed as a discrete structural realization of the same bounded multiplicative principle, depending on the statistical structure of the κ_k and of B .

A different possibility is a self-normalizing stochastic field, motivated by the regenerative nature of spike propagation itself. In such a picture, local fluctuations in excitability or conduction state need not accumulate freely because each active segment

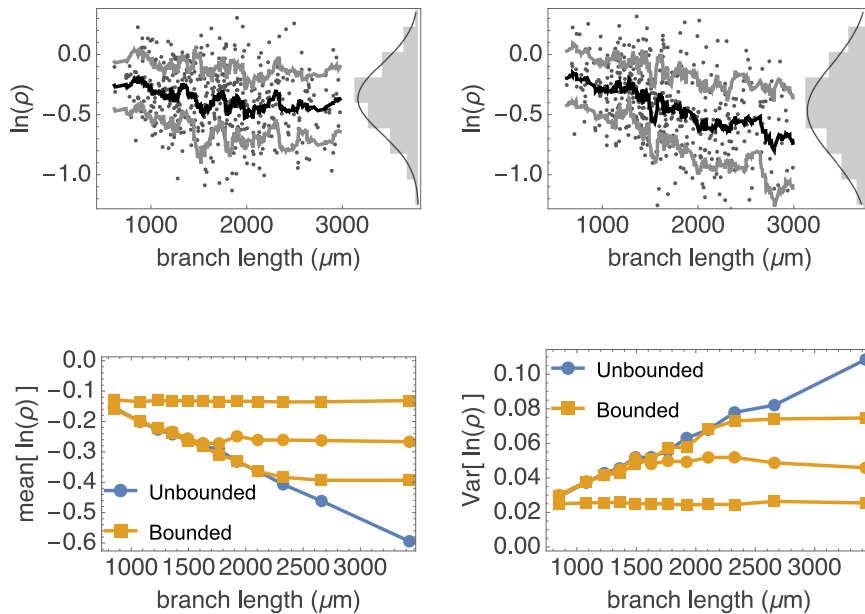


Figure 2. Monte Carlo comparison of bounded and unbounded multiplicative slowdown. Synthetic slowdown ratios $\rho = v_{\text{end}}/v_{\text{start}}$ are simulated under two multiplicative models that differ only in the effective number of compounding steps along a branch: unbounded depth increases with branch length, whereas bounded depth saturates beyond a saturation scale. Top: simulated $\ln \rho$ versus total soma-to-terminal branch length for the bounded (left, saturation scale is 1,600 μm) and unbounded (right) models, with binned central trends and marginal histograms of $\ln \rho$. Bottom: length-binned estimates of $\mathbb{E}[\ln \rho]$ (left) and $\text{Var}[\ln \rho]$ (right). In the bounded model, both the mean and variance approach a plateau whose onset depends on the saturation scale {800, 1600, 2400} μm . In the unbounded model, they grow with length. Simulation details and parameter calibration are given in the [supplemental methods](#).

partially restores the propagating event toward a local effective set point. This type of mechanism naturally limits the growth of $\text{Var}[\ln \rho]$ with distance and therefore provides a plausible account of constrained spread. However, it is orthogonal to the observed mean slowdown: to obtain $\rho \approx 0.7$, one must additionally assume that the effective target state is itself biased below the proximal reference value. Thus, self-normalization can explain bounded dispersion but not the observed central tendency unless supplemented by an additional distal or state-dependent bias.

Another possibility is a correlated multiplicative field, in which slowdown reflects continuously distributed local heterogeneity with finite spatial correlation length, so that nearby segments contribute nonindependently rather than as isolated weak points. Such correlations can limit the effective growth of $\text{Var}[\ln \rho]$ with distance by reducing the number of effectively independent multiplicative factors. In this respect, the correlated-field picture differs from the stochastic weak-link picture in ontology: the latter is event based and governed by the number and strength of discrete local slowing events, whereas the former is distributed and governed by correlated spatial variation in local propagation determinants. However, although such a field can account for constrained spread, it does not by itself explain why the central tendency should remain stably near $\rho \approx 0.7$.

The alternatives clarify the constraint imposed by the data. Unbounded multiplicative accumulation naturally yields right-skewed ρ but predicts growth

of $\text{Var}[\ln \rho]$ with length. Discrete weak-link or branch-point models can generate slowdown but predict localized jumps unless additional smoothing assumptions are introduced. Self-normalizing stochastic fields can stabilize dispersion but do not, by themselves, explain the mean bias toward $\rho \approx 0.7$. Correlated multiplicative fields can smooth local velocity profiles but require additional structure to account for both the stable mean and stable dispersion. Thus, the distinctive empirical target is the joint occurrence of right-skewed slowdown, mean bias, and length-invariant dispersion.

INTERPRETATION AND DISCRIMINATING PREDICTIONS

Products of random factors arise broadly in transport through heterogeneous media, growth processes, and cascade models. In unbounded multiplicative accumulation, the dispersion of $\ln \rho$ is expected to grow with multiplicative depth. Here, a living excitable medium is analyzed, axons in which $\rho = v_{\text{end}}/v_{\text{start}}$ is approximately log normal, yet the mean and dispersion of $\ln \rho$ show no evidence of dependence on axonal length over the full range examined. This combination of multiplicative form and bounded mean and dispersion constrains the admissible coarse-grained dynamical models, pointing to mechanisms that saturate the effective multiplicative depth rather than allowing unbounded accumulation.

The main claim is that axonal slowdown is captured, at the level of distributional structure, by a

bounded multiplicative process modulated by distal boundary conditions:

$$\rho = \eta(\Lambda) \prod_{i=1}^{\eta_{\text{eff}}(\Lambda)} \kappa_i,$$

with $\eta(\Lambda) = \eta_{\text{structure}}(\Lambda) \eta_{\text{kinetic}}(\Lambda)$. The log normality and length invariance of ρ are empirical invariances; the bounded multiplicative form is proposed as a minimal arrangement that satisfies them simultaneously. The argument therefore runs from invariances to a constrained class of mechanisms, not from a distributional fit to a specific microscopic cause. The measured slowdown ratio constrains the aggregate form of the process, but it does not identify each local microscopic contribution. Assigning individual multiplicative factors κ_i to specific biophysical entities, such as diameter changes, branch points, channel-density gradients, or local excitability losses, would therefore add apparent precision without additional empirical constraint. Likewise, biophysical simulations that instantiate particular choices of the κ_i and of $\eta(\Lambda)$ are possible, but they would be illustrative rather than discriminating: many distinct microscopic implementations can realize the same bounded multiplicative law (see, for instance, [Figure S1](#)).

At this resolution, the aim is to distinguish distal boundary regimes and their experimentally visible signatures rather than enumerate microscopic contributors. This motivates three experimentally accessible predictions designed to discriminate between structural load/impedance and kinetic reserve depletion. The proposed manipulations cannot furnish a perfectly orthogonal decomposition. Rather, they create conditions under which one contribution is expected to carry greater weight than the other.

First, if orthodromic and antidromic spikes are initiated simultaneously along the same branch, their approach to collision creates a transient effective termination zone in which the wavefronts annihilate. Within the decomposition $\eta(\Lambda) = \eta_{\text{structure}}(\Lambda) \eta_{\text{kinetic}}(\Lambda)$, this paradigm should preferentially reflect η_{kinetic} : as the fronts converge, regenerative margin should be transiently reduced by availability-dependent kinetics (for example, inactivation- and accommodation-like depletion of available excitatory sodium conductance), yielding measurable slowing relative to unidirectional propagation under otherwise identical conditions. Detection of collision-associated slowing, with minimal manipulation of impedance, would therefore support a nonnegligible contribution from $\eta_{\text{kinetic}}(\Lambda)$. Conversely, the absence of such slowing would be more consistent with a predominantly

structural contribution, $\eta_{\text{structure}}(\Lambda)$, anchored to terminal architecture rather than collision-induced reserve depletion. Operationally, the comparison is between local velocity profiles measured during unidirectional propagation and during controlled orthodromic-antidromic collision along the same branch; failure to detect any collision-specific slowing, despite sufficient temporal and spatial resolution, would argue against a substantial kinetic-reserve contribution.

Second, the mean of the $\ln \rho$ distribution should increase (or, at least, not decrease) when spikes are initiated at the far end of the axon and propagate to the soma. Here, ρ is defined, as above, by $\rho = v_{\text{end}}/v_{\text{start}}$, with “start” denoting the distal initiation site and “end” the soma. This paradigm provides a comparatively cleaner window onto $\eta_{\text{structure}}$, as the time-independent terminal architecture is probed without deliberately engaging the kinetic reserve. A terminal-anchored bias is, by construction, specific to the end at which it sits, so reversing the direction of initiation should lift it. Operationally, the key measurement is the mean of $\ln \rho$ under orthodromic versus antidromic propagation in the same branch; a nondecrease in the antidromic mean supports a terminal-anchored structural bias, whereas a decrease argues against it.

Third, for branches whose physical length substantially exceeds the effective distal coupling length Λ (measured from the sealed end toward the soma), the velocity profile is predicted to contain an extended proximal segment with approximately constant, or only weakly declining, velocity, followed by a pronounced distal deceleration confined to a terminal region of order Λ . In very long axons, such a “free-running” initial segment should therefore be observable in length-velocity profiles, with the bounded multiplicative slowdown emerging only as the spike enters the sealed-end zone. Because the electrode field in the device used in Cohen and Marom¹² spans only ca. 2×4 mm, the data set does not include axons long enough to evaluate this asymptotic prediction; testing it would require a recording modality capable of probing substantially longer trajectories. Operationally, the test is whether $\text{Var}[\ln \rho]$ saturates with length once trajectories exceed Λ ; continued growth of $\text{Var}[\ln \rho]$ with branch length would falsify the bounded-depth interpretation and favor unbounded accumulation.

Together, these predictions specify what should be manipulated and measured to discriminate among distal boundary regimes (load/impedance versus excitability reserve) that can otherwise generate the same bounded slowdown statistics. The same statistics admit multiple realizations: passive impedance transformations, branch geometry, and fixed

channel-density gradients enter as components of $\eta_{\text{structure}}(\Lambda)$, while history- and voltage-dependent reductions in excitability reserve enter through $\eta_{\text{kinetic}}(\Lambda)$, and the measured ρ does not by itself separate them. This underdetermination is what the predictions are designed to break.

The preceding argument concerns axons that share similar boundary conditions. Where those conditions differ fundamentally, including myelinated, demyelinating, or otherwise pathologically altered axons, the framework predicts different slowdown statistics, and such preparations would constitute important tests of the framework.

The length-invariance of the ρ distribution has a direct functional consequence: longer branches conduct faster on average than shorter ones because the terminal slowdown zone of scale Λ constitutes a smaller fraction of their total trajectory, reducing the dispersion of arrival times across branches of different lengths. The energetic implication follows: achieving this reduced arrival time dispersion does not require coordinated fine-tuning of channel density, membrane geometry, or kinetic parameters along the entire axonal trajectory; the boundary condition at scale Λ is sufficient.

Overall, the results place axonal conduction slowdown within a class of self-normalizing physiological processes in which local variability is preserved while global divergence is constrained. Earlier work has shown that excitability can be stabilized against structural and kinetic heterogeneity through activity-dependent normalization mechanisms.^{15–18} The present findings extend this logic to spatial propagation: local geometric and kinetic influences may compound proportionally along an axonal trajectory, while boundary conditions constrain the net effect expressed at the endpoint. The axonal example may therefore be more than a special case. It may exemplify a broader physiological organization in which proportional local influences accumulate without allowing variability to diverge. Robust function emerges through constrained variability: a relational balance between local heterogeneity and global boundary conditions.

DATA AVAILABILITY

Wolfram Language scripts used to generate the figures are provided as [Data S1 \(Figure S1\)](#), [Data S2 \(Figure S2\)](#), and [Data S3 \(Figure 2\)](#), and are also deposited at Mendeley Data: <https://doi.org/10.17632/k42gs7cxcw.1>.

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AUTHOR CONTRIBUTIONS

S.M. designed the research, performed the research, analyzed the data, and wrote the paper.

DECLARATION OF INTERESTS

The author declares no competing interests.

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.bpr.2026.100277>.

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Supplemental information

**Bounded multiplicative dynamics
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Supplementary Material for “Bounded multiplicative dynamics govern axonal conduction slowdown”

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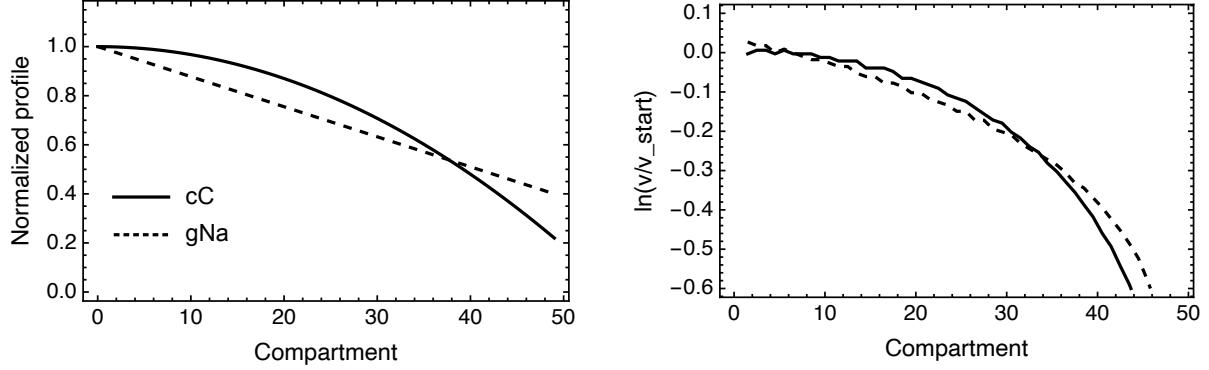


Figure S1. Structural realizations of terminal slowdown in a propagated Hodgkin–Huxley cable. Two single-run simulations of a 50-compartment Hodgkin–Huxley cable show that distinct fixed structural gradients can yield terminal slowdown ratios within the empirical range. **Left:** normalized spatial profiles. Solid line: effective axial coupling coefficient cC/cC_0 (run A), decreasing distally and representing reduced source-to-load coupling. Dashed line: peak sodium conductance $\hat{g}_{Na}/\hat{g}_{Na,0}$ (run B), decreasing distally with axial coupling held uniform. **Right:** local conduction velocity expressed as $\ln(v/v_{start})$. Run A gives $\ln \rho \approx -0.44$; run B gives $\ln \rho \approx -0.38$. The simulations are illustrative rather than identifying: both gradients are time-independent structural realizations of $\eta_{structure}(\Lambda)$, and the similarity of ρ shows that the slowdown statistic does not uniquely determine the microscopic mechanism. Code is provided in the accompanying Mathematica script SuppFigS1.wl

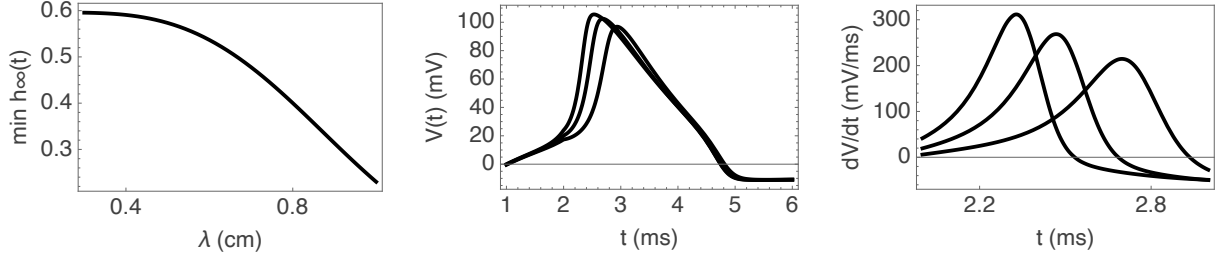


Figure S2. Availability-based realization of an excitability-reserve bias and its spike-shape signature. The figure provides a computable proxy for a deterministic excitability-reserve contribution to $\eta(\Lambda)$ and illustrates an experimentally accessible correlate of reduced regenerative reserve. It is not used in the bounded-depth derivation. **Left:** Effective minimum sodium-channel availability, $h_{\text{eff}}(\lambda) = \min_{t \in [0, T]} h_{\infty}(V_{\text{end}}(t; \lambda))$, computed from the end voltage produced by an illustrative passive end-sensing construction in which a clamped depolarization waveform $V_c(t)$ approaches a sealed end at speed v_{app} ; increasing λ reduces attenuation and raises $V_{\text{end}}(t)$, thereby lowering h_{eff} under depolarizing conditions. **Middle:** Single-compartment Hodgkin–Huxley spikes evoked by identical stimulus current, with all parameters held fixed and only $h(0)$ set to three representative values spanning the range of $h_{\text{eff}}(\lambda)$. **Right:** Corresponding upstroke traces dV/dt , showing a monotone reduction in $\max(dV/dt)$ with reduced initial availability. Code is provided in the accompanying Mathematica script SuppFigS2.wl

Supplementary Methods

Monte Carlo Illustration of Bounded versus Unbounded Multiplicative Depth

Synthetic slowdown ratios were generated by a log-additive multiplicative process,

$$\ln \rho(L_s) = \sum_{i=1}^{n(L_s)} \zeta_i, \quad \zeta_i \sim \mathcal{N}(\mu, \sigma^2) \text{ i.i.d.}, \quad (1)$$

where L_s denotes the total soma-to-terminal branch length, not the distance-from-tip coordinate used elsewhere. (In this construction the deterministic boundary factor is set to $\eta(\Lambda) = 1$, so the mean slowdown is carried entirely by the accumulated per-step mean μ rather than by $\ln \eta(\Lambda)$; this is a calibration convenience and does not affect the variance behavior the figure illustrates.) Branch lengths L_s were drawn from a truncated log-normal distribution supported on 600–4570 μm , chosen to approximate the empirical length distribution. To define a length-dependent “multiplicative depth” $n(L_s)$, a coarse-graining length ℓ is introduced, converting a continuous length into an integer step count. The length ℓ is a modeling convention, namely a resolution scale for $n(L_s)$, not a claim of a special anatomical segment length. In Fig. 2, $\ell = 200 \mu\text{m}$ is set as a convenient order-of-magnitude discretization.

The unbounded model assumes the effective depth increases with length,

$$n_{\text{unb}}(L_s) = \left\lceil \frac{L_s}{\ell} \right\rceil, \quad (2)$$

whereas the bounded model caps the depth at a finite saturation value,

$$n_{\text{bnd}}(L_s) = \begin{cases} \left\lceil \frac{L_s}{\ell} \right\rceil, & \text{if } \left\lceil \frac{L_s}{\ell} \right\rceil \leq n_\infty, \\ n_\infty, & \text{if } \left\lceil \frac{L_s}{\ell} \right\rceil > n_\infty. \end{cases} \quad (3)$$

Here n_∞ is the finite asymptotic effective depth, namely the maximal number of independent multiplicative factors permitted by the model at large L_s . Because ℓ is a coarse-graining convention, the associated saturation occurs on the effective length scale set by $n_\infty \ell$; this scale is part of the model representation and is not uniquely identified as a physiological length.

Per-step parameters were chosen so that, in the bounded model at saturation, $n(L_s) = n_\infty$, the marginal distribution matches the empirical branch-level full-set fit $\ln \rho \sim \mathcal{N}(\hat{\mu}_\rho, \hat{\sigma}_\rho^2)$ ¹. Since at saturation

$$\mathbb{E}[\ln \rho] = n_\infty \mu, \quad \text{Var}[\ln \rho] = n_\infty \sigma^2, \quad (4)$$

matching gives

$$\mu = \frac{\hat{\mu}_\rho}{n_\infty}, \quad \sigma = \frac{\hat{\sigma}_\rho}{\sqrt{n_\infty}}. \quad (5)$$

For Fig. 2 (top panels), $n_\infty = 8$ was used as a representative finite cap, yielding $\mu = -0.04929$ and $\sigma = 0.09648$ from $\hat{\mu}_\rho = -0.3943$ and $\hat{\sigma}_\rho = 0.2729$, so $\sigma^2 = 9.31 \times 10^{-3}$.

¹L. D. Cohen, S. Marom, Distributional invariance and proportional scaling in axonal conduction, Biophysical Journal 125 (7) (2026) 1660–1666.

For each model, $N = 2 \times 10^4$ branches were simulated. Scatter plots show $\ln \rho$ versus L_s , with equal-count binned central trends and marginal histograms pooling $\ln \rho$ across branches. Bottom panels show equal-count length-bin estimates of $\mathbb{E}[\ln \rho]$ and $\text{Var}[\ln \rho]$, compared to the expectations $\mathbb{E}[\ln \rho(L_s)] = n(L_s)\mu$ and $\text{Var}[\ln \rho(L_s)] = n(L_s)\sigma^2$. For visual clarity, Fig. 2 displays only $L_s \leq 3.3$ mm.

Code is provided in the accompanying Mathematica script `MonteCarloFig2.wl`